

features of dynamical behavior (convergence to steady states, periodic orbits, chaos, etc.). A central question, thus, is that of understanding the long-time behavior of solutions. In mathematical terms, and using standard chemical kinetics modeling, this problem may be translated into the study of the set of possible limit points (the ω -limit set) of the solutions of a system of ordinary differential equations.

1.1. Robustness

A distinguishing feature of this study in the context of cell biology, in contrast to more established areas of applied mathematics and engineering, is the very large degree of uncertainty inherent in models of cellular biochemical networks. This uncertainty is due to environmental fluctuations, and variability among different cells of the same type, as well as, from a mathematical analysis perspective, the difficulty of measuring the relevant model parameters (kinetic constants, cooperativity indices, and many others) and thus the challenge to obtain a precise model. Thus, it is imperative to develop tools that are ‘robust’ in the sense of being able to provide useful conclusions based only upon information regarding the *qualitative* features of the network, and not the precise values of parameters or even the forms of reactions. Of course, this goal is often unachievable, since dynamical behavior may be subject to phase transitions (bifurcation phenomena) which are critically dependent on parameter values.

Nevertheless, and surprisingly, research by many, notably by Clarke [10], Horn and Jackson [29,30], Feinberg [18–20], and many others in the context of complex balancing and deficiency theory, and by Hirsch and Smith [41,26] and many others including the present authors [2,17,3,9] in the context of monotone systems, has resulted in the identification of rich classes of chemical network structures for which such robust analysis is indeed possible. In this paper, we present yet another approach to the robust analysis of dynamical properties of biochemical networks, and apply our approach in particular to the analysis of persistence in chemical networks modeled by ordinary differential equations. Our approach to study persistence is based on the formalism and basic concepts of the theory of Petri nets. Using these techniques, our main results provide conditions (some necessary, and some sufficient) to test persistence. We then apply these conditions to obtain fairly tight characterizations in non-trivial examples arising from the current molecular biology literature.

1.2. Persistence

Persistence is the property that, *if every species is present at the start of the reaction, no species will tend to be eliminated in the course of the reaction*. Mathematically, this property can be equivalently expressed as the requirement that the ω -limit set of any trajectory which starts in the interior of the positive orthant (all concentrations positive) does not intersect the boundary of the positive orthant (more precise definitions are given below). Persistence can be interpreted as non-extinction: if the concentration of a species would approach zero in the continuous differential equation model, this means, in practical terms, that it would completely disappear in finite time, since the true system is discrete and stochastic. Thus, one of the most basic questions that one may ask about a chemical reaction network is if persistence holds for that network. Also from a purely mathematical perspective persistence is very important, because it may be used in conjunction with other

techniques in order to guarantee convergence of solutions to equilibria. For example, if a strictly decreasing Lyapunov function exists on the interior of the positive orthant (see e.g. [29,30,18–20,42] for classes of networks where this can be guaranteed), persistence allows such a conclusion.

An obvious example of a non-persistent chemical reaction is a simple irreversible conversion $A \rightarrow B$ of a species A into a species B ; in this example, the chemical A empties out, that is, its time-dependent concentration approaches zero as $t \rightarrow \infty$. This is obvious, but for complex networks determining persistence, or lack thereof, is, in general, an extremely difficult mathematical problem. In fact, the study of persistence is a classical one in the (mathematically) related field of population biology, where species correspond to individuals of different types instead of chemical units; see for example [22,7] and much other foundational work by Waltman. (To be precise, what we call ‘persistence’ coincides with the usage in the above references, and is also sometimes called ‘strong persistence,’ at least when all solutions are bounded, a condition that we will assume in most of our main results, and which is automatically satisfied in most examples. Also, we note that a stronger notion, ‘uniform’ persistence, is used to describe the situation where all solutions are eventually bounded away from the boundary, uniformly on initial conditions, see [8,44].) Most dynamical systems work on persistence imposes conditions ruling out phenomena such as heteroclinic cycles on the boundary of the positive orthant, and requiring that the unstable manifolds of boundary equilibria should intersect the interior, and more generally studying the chain-recurrence structure of attractors, see e.g. [27].

1.3. Petri nets

Basic ideas introduced by Carl Adam Petri in 1962 [38] led to the notion of a *Petri net*, also called a place/transition nets, and they constitute a popular mathematical and graphical modeling tool used for concurrent systems modeling [37,47]. Our modeling of chemical reaction networks using Petri net formalism is not in itself a new idea: there have been many works, at least since [39], which have dealt with biochemical applications of Petri nets, in particular in the context of metabolic pathways, see e.g. [23,28,32,35,36,46]. In this paper, we associate both a Petri net and a system of differential equations to a chemical reaction network. The latter describes the behavior of the concentrations of the chemicals in the network. We intend to draw conclusions about the asymptotic behavior of the solutions of the system of differential equations, based on the graphical and algebraic properties of the associated Petri net. This is very related to open questions which have been raised in recent works by Gilbert and Heiner as well as Silva and Recalde, [24,40], where a similar point of view is taken, of either complementing discrete analysis by means of continuous techniques or integrating the two approaches for a deeper understanding (see [16] for an introduction to continuous Petri nets).

Although we do not use any results from Petri net theory, we employ several concepts (siphons, P-semiflows, etc.), borrowed from that formalism and introduced as needed, in order to formulate new, powerful, and verifiable conditions for persistence and related dynamical properties.

1.4. Application to a common motif in systems biology

In molecular systems biology research, certain ‘motifs’ or subsystems appear repeatedly, and have been the subject of much recent research. One of the most common ones is that in which

and

$$\{S_0, S_1, E^\star, ES_0, ES_1, FS_2, FS_1, ES_0^\star, ES_1^\star\}.$$

Each one of them is the support of a P-semiflow, and there are five conservation laws:

$$\begin{aligned} E + ES_0 + ES_1 &= \text{const}_1, \\ F + FS_2 + FS_1 &= \text{const}_2, \\ F^\star + FS_2^\star + FS_1^\star &= \text{const}_3, \\ S_0^\star + S_1^\star + S_2^\star + ES_0^\star + ES_1^\star + FS_2^\star + FS_1^\star &= \text{const}_4, \end{aligned}$$

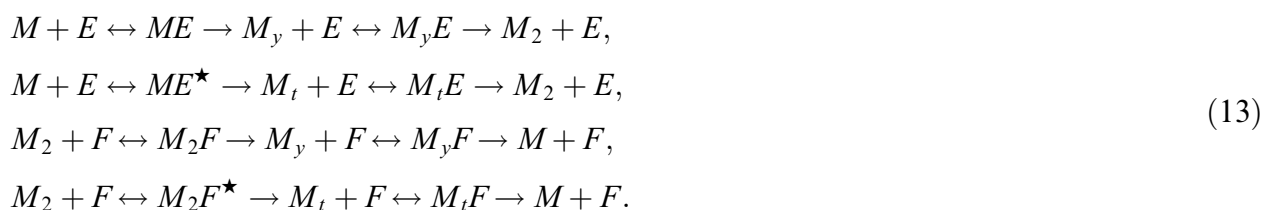
and

$$S_0 + S_1 + E^\star + ES_0 + ES_1 + FS_2 + FS_1 + ES_0^\star + ES_1^\star = \text{const}_5.$$

As in the previous example, the network is conservative since the sum of these conservation laws is also a conservation law. Therefore the overall network is persistent, by virtue of [Theorem 2](#). It is worth pointing out that the number of minimal siphons of a network may grow even exponentially with the size of the network. For large scale networks, it becomes therefore crucial to obtain algorithms for the determination of all minimal siphons in order to automatically check the assumptions of [Theorem 2](#). The paper [\[13\]](#) presents one such algorithm, together with some numerical and analytical results dealing with problem complexity.

6.3. Example 3

An alternative mechanism for dual phosphorylation in MAPK cascades, considered in [\[34\]](#), differs from the previous ones in that it becomes important at which sites the two phosphorylations occur. (These take place at two different sites, a threonine and a tyrosine residue). The corresponding network can be modeled as follows:



See [Fig. 4](#) for the corresponding Petri net. This network is consistent. Indeed, $\Gamma v = 0$ for the same v as in the previous example. Moreover it admits three siphons of minimal support:

$$\begin{aligned} \{E, ME, ME^\star, M_yE, M_tE\}, \\ \{F, M_yF, M_tF, M_2F, M_2F^\star\}, \end{aligned}$$

and

$$\{M, ME, ME^\star, M_y, M_t, M_yE, M_tE, M_2, M_2F, M_2F^\star, M_tF, M_yF\}.$$

Petri Nets, and actually a complete characterization of liveness if the net is a ‘Free Choice Petri Net’ (this is known as Commoner’s Theorem, [25] and [11]):

Theorem 3. *Consider a conservative Petri net satisfying the following assumption:*

each (minimal) siphon contains a non-empty trap.

Then, the PN is alive.

Notice the similarity between the assumptions and conclusions in Theorem 2 and in Theorem 3. There are some subtle differences, however. Traps for Petri nets enjoy the following invariance property: if a trap is non-empty at time zero (meaning that at least one of its places has tokens), then the trap is non-empty at all future times. In contrast, in a continuous set-up (when tokens are not integer quantities but may take any real value), satisfaction of the siphon-trap property does not prevent (in general) concentrations of species from decaying to zero asymptotically. This is why we needed a strengthened assumption 2, and asked that each siphon contains the support of a P-semiflow (which is always, trivially, also a trap). In other words, in a continuous set-up the notion of a trap loses much of its appeal, since one may conceive situations in which molecules are pumped into the trap at a rate which is lower than the rate at which they are extracted from it, so that, in the limit, the trap can be emptied out even though it was initially full. A similar situation never occurs in a discrete set-up since, whenever a reaction occurs, at least one molecule will be left inside the trap.

8. Conclusions

In ecology, persistence is the property of an ecosystem to asymptotically preserve non-zero populations of all the species which are present at the initial time. In the present paper we obtain both necessary and sufficient conditions for persistence in chemical reaction networks under a general monotonicity assumption for the reaction rates. The conditions are stated in terms of graphical and algebraic properties of Petri nets which are associated to the chemical reaction network. In a subsequent paper we will present tighter results for networks in which all reaction rates are of mass action type. The result presented here may also serve as a preliminary step towards the construction of a systematic input/output theory for chemical reaction networks, by allowing systems with inflows and outflows.

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